Local Burden and Demographic Disparities in Alzheimer's Disease and Related Dementias: A Generalizability Analysis of Stanford ADRC Clinical Core Data

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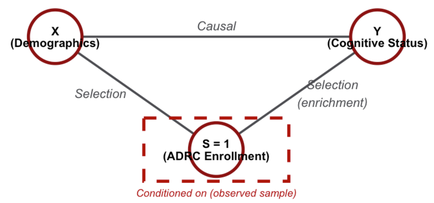
Background

- **Alzheimer's disease (AD) and AD-related dementias (AD/ADRD)** affect over 6 million US adults. National studies estimate ~9% AD prevalence (Dhana et al. 2023) and ~10% ADRD prevalence (Matthews et al. 2019) among adults 65+, but local estimates remain limited.
- **National models often lack estimates for Asian adults** – a critical gap for the Bay Area, where Asian residents represent a substantial share of the older adult population.
- **Local estimates** are needed for resource allocation and caregiver support planning, but most existing estimates come from national models that may not reflect regional demographics.
- **Alzheimer's Disease Research Centers (ADRCs)** provide rich clinical data but over-enroll people with cognitive impairment. This **clinical enrichment** complicates generalization of ADRC findings.



Research Questions

1. What is the estimated local prevalence of AD/ADRD among adults aged 65+ in the ADRC's three-county catchment area?
2. Are there disparities in cognitive outcomes across racial/ethnic groups after demographic standardization?
3. How does clinical enrichment shape the ADRC sample relative to the target population?

Data & Methods

- **Data:** We used data from the Stanford ADRC (529 adults 65+, NACC UDS v3, 2015–2024), California BRFSS (2,150 adults 65+ in Alameda, San Mateo, and Santa Clara counties, 2015–2024), and published coefficients from Dhana et al. (2023) and Matthews et al. (2019).
- **Q1 – Local AD/ADRD Prevalence:** Dhana et al. published logistic regression coefficients for AD prevalence, but their model lacks an Asian category. We constructed a hybrid model:

$$\text{logit}(P(\text{AD}_i)) = \hat{\alpha}^D + \hat{\beta}^D_{\text{age}} x_{\text{age}} + \hat{\beta}^D_{\text{sex}} x_{\text{sex}} + \hat{\beta}^D_{\text{edu}} x_{\text{edu}} + \hat{\beta}^A_{\text{race}} x_{\text{race}}$$

  where D denotes published CHAP coefficients and $\hat{\beta}^A_{\text{race}}$ are log-odds ratios for AD by race/ethnicity (vs. White) from the Stanford ADRC. We verified transportability of these ORs by testing for variation in clinical enrichment across groups. We applied the model to individual-level BRFSS data with survey weights. For ADRD, we used Matthews et al. (2019) rates via indirect standardization.
- **Q2 – Disparities (AIPSW):** We analyzed AD, ADRD, and mild cognitive impairment (MCI) as separate outcomes. We estimated prevalence ratios using augmented inverse probability of selection weighting (Dahabreh et al. 2019), a doubly robust estimator combining outcome models with stabilized inverse odds of selection weights. We estimated variance via stratified bootstrap (1,000 reps × 40 imputations, Rubin's rules).
- **Q3 – Enrichment:** We characterized ADRC referral pathways and compared ADRC-derived odds ratios to published population-based estimates to verify that enrichment is approximately constant across groups – a key assumption for transportability of the disparity analysis.

Results

Figure 1. Estimated AD/ADRD prevalence by race/ethnicity

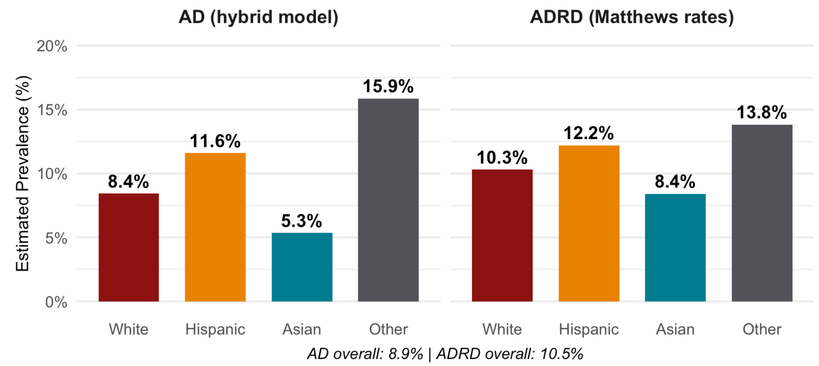


Figure 2. AIPSW prevalence ratios by race/ethnicity

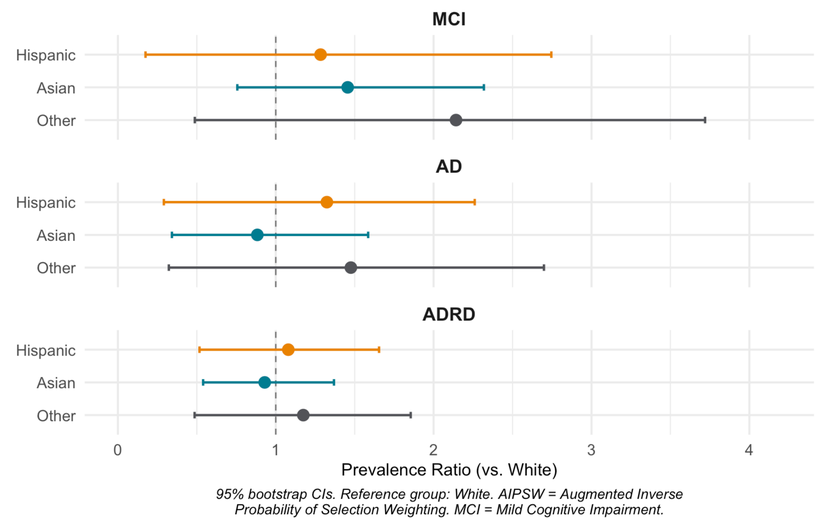


Figure 3. Cognitive impairment by referral pathway

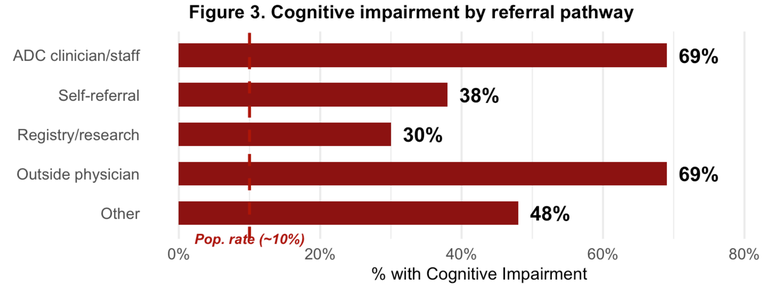
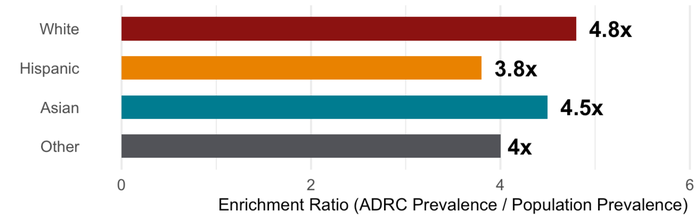


Figure 4. Clinical enrichment ratio by race/ethnicity



Conclusions & Future Directions

- We estimate AD prevalence at **8.9%** among adults 65+ in the three-county catchment, highest among Hispanic (11.6%) and lowest among Asian older adults (5.3%).
- We estimate ADRD prevalence at **10.5%** overall, highest among White (15.9%) and Hispanic (13.8%) older adults.
- Clinical enrichment is pervasive across all referral pathways. Demographic reweighting **does not enable population prevalence estimation** from ADRC data alone.
- However, approximately constant enrichment across groups supports the use of reweighting to estimate **racial and ethnic disparities**.
- **Black adults are underrepresented** in the ADRC (n=12 at 65+). Expanding recruitment is a priority.

Limitations

- The hybrid model assumes constant enrichment across groups. We tested this empirically (3.8x–4.8x) but with limited precision.
- The Hispanic (n=58) and Other (n=18) ADRC subgroups yield wide confidence intervals.
- The ADRC's etiological composition (~45% Lewy body disease) reflects Stanford's research focus and may not generalize.
- We collapsed Black participants into Other due to small sample size (n=12), precluding Black-White disparity estimation.
- BRFSS excludes institutionalized adults, among whom ADRD prevalence is higher.
- Dhana et al. estimated the CHAP coefficients in a Chicago population; these may not transfer perfectly to the Bay Area.
- The NACC UDS does not capture several covariates used in prior generalizability analyses (income, self-rated health, functional limitations).

Future Directions

- Expand recruitment of underrepresented groups, particularly Black older adults.
- Disaggregate the Asian category given known heterogeneity across subgroups (Mayeda et al. 2016).
- Supplement NACC data with site-level collection of covariates (e.g., income, functional limitations) not captured in the UDS.
- Replicate this generalizability framework across other ADRCs to assess consistency of enrichment patterns.

References

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